

Guideline for the laboratory diagnosis of iron deficiency in adults (excluding pregnancy) and children

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Methodology

This Good Practice Paper was compiled according to the British Society for Haematology (BSH) process at: <https://b-s-h.org.uk/media/16732/bsh-guidance-development-process-dec-5-18.pdf>. The BSH produces 'Good Practice Papers' to recommend good practice in areas where there is a limited evidence base but for which a degree of consensus or uniformity is likely to be beneficial to patient care. The Grading of Recommendations Assessment, Development and Evaluation (GRADE) nomenclature was used to evaluate levels of evidence and to assess the strength of recommendations. The GRADE criteria can be found at <http://www.gradeworkinggroup.org>.

Literature review details

Up to June 2020 MEDLINE and PUBMED were searched systematically for publications in English from 1966 to 2020, with additional *ad hoc* searches up to June 2021 for any new evidence using key words: iron deficiency, anaemia and each of the parameters discussed.

Review of the manuscript

Review of the manuscript was performed by the BSH Guidelines Committee General Haematology Task Force, the BSH Guidelines Committee and the General Haematology sounding board of the BSH. It was made available on the members' section of the BSH website for comment.

Introduction

The laboratory diagnosis of iron deficiency is difficult because iron homeostasis is dynamic. No single test can

provide an accurate assessment of iron absorption, transport, storage, and utilisation. The different assays available to assess iron and its stores will each be discussed, and recommendations made pertinent to practice within the UK. Iron metabolism in adults and children can be considered equivalent and these recommendations are applicable to both paediatric and adult practice.

Iron deficiency is the most common cause of anaemia worldwide. The World Health Organization (WHO) defines anaemia as a condition in which the number of red cells (and consequently their oxygen-carrying capacity) is insufficient to meet the body's physiological needs.¹ In addition to this there is a recognised definition of haemoglobin concentration below age- and gender- specific thresholds: <130 g/l for adult men, <120 g/l for adult non-pregnant women and paediatric values that start at <110 g/l for ages 6–59 months and increase with age.¹ The prevalence of iron deficiency anaemia in the UK population ranges between 0% and 6% according to age and sex, higher rates being recognised in certain demographic groups such as menstruating women and adults aged >85 years.² Prevalence of iron deficiency in the absence of anaemia is not well documented.

Laboratory tests should be performed in the context of a detailed history and clinical examination with an awareness of how results will direct onward investigation and management based on national gastrointestinal and gynaecology guidelines and local pathways within individual healthcare settings. This good practice paper should benefit all healthcare professionals investigating paediatric and non-pregnant adult patients with suspected iron deficiency. It supplements existing BSH guidelines for the specific situations of diagnosing functional iron deficiency and the diagnosis and management of iron deficiency in pregnancy.^{3,4} A suggested algorithm for the investigation of iron deficiency anaemia is included below (Fig 1).

Calculation of red cell values haematocrit (Hct), mean corpuscular volume (MCV), mean cell haemoglobin (MCH), mean corpuscular haemoglobin concentration (MCHC) and red cell distribution width (RDW) can be found in Dacie and Lewis Practical Haematology.⁵

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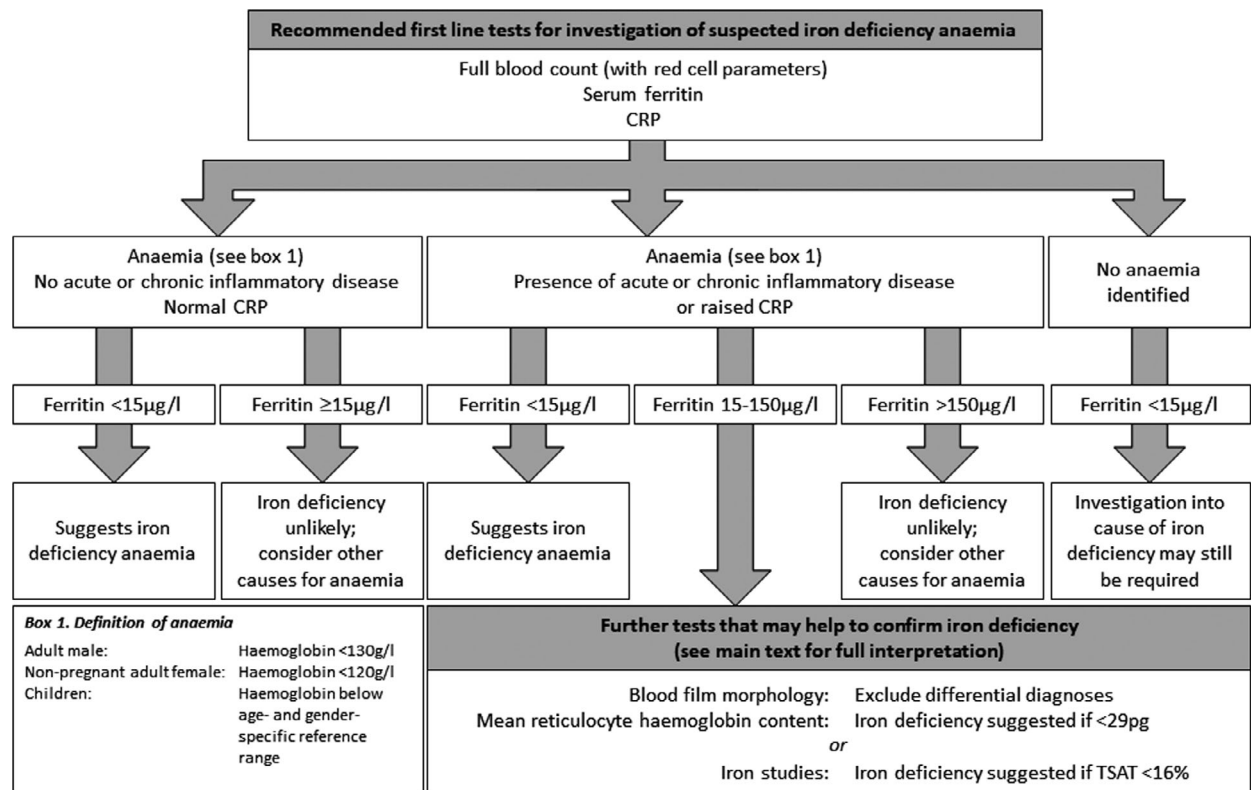


Fig 1. Example algorithm for diagnosis of iron deficiency anaemia with locally adapted values. Tests not currently recommended for the further investigation of iron deficiency: zinc protoporphyrin (ZPP), soluble transferrin receptor (sTfR), hepcidin. Key: full blood count (FBC), C-reactive protein (CRP), transferrin saturation (TSAT).

Haemoglobin concentration and red cell parameters (Hct, MCV, MCH, MCHC, RDW)

The full blood count (FBC) will confirm anaemia if haemoglobin (Hb) is below the laboratory reference range. Modern laboratory spectrophotometric (or colorimetric) measurement of Hb is accurate and reliable. Hb and its related parameter Hct do not provide information with regard to iron status. At population level there is significant overlap in the distribution of Hb between iron-replete and -deficient individuals.⁶

Automated FBC analysis includes the validated red cell parameters MCV, MCH and MCHC, as well as RDW, although in the UK this latter index is not yet quality assured. Each can be reduced by an iron deficient state, but not to a consistent enough degree for diagnostic clarity. Iron deficiency is classically associated with microcytic hypochromic anaemia, but low MCV, MCH or MCHC can be seen in other conditions, notably the thalassaemias. MCV can be normal in up to 40% of unselected iron-deficient patients and also in states of mixed haematinic deficiency.⁷ MCV is well known to be affected by pre-analytical variables such as sample temperature and storage times.^{8,9} MCV, MCHC and RDW cannot be directly compared between analysers. Although the MCH retains accuracy and stability between analysers and is somewhat less affected by pre-analytical

variables, it is not specific for iron deficiency. None of these parameters alone is able to predict response to oral iron.¹⁰⁻¹²

Recommendations

- If thalassaemia is not present, then any of MCV, MCHC or MCH below the normal range could suggest iron deficiency (2C)
- Normal red cell indices do not exclude iron deficiency as a cause of anaemia, so further investigation may still be required if there is a high level of clinical suspicion (1B)

Advanced red cell indices [percentage of hypochromic cells (%HRC), reticulocyte mean Hb content, reticulocyte Hb equivalent)

Erythrocytes contain approximately half the body iron at any one time.¹³ The red cell indices offered by more recent analysers can be predictive in diagnosing iron deficiency and response to iron (Table I). These advanced parameters include %HRC and the reticulocyte mean Hb content. Methods and nomenclature for these indices vary by analyser, but a high %HRC or low reticulocyte mean Hb content in persons without thalassaemia indicates that iron is not

adequately available to the newly developed red cells. This can be due to iron deficiency or iron restriction.¹⁰

The %HRC is defined as the proportion of erythrocytes with cellular Hb of <280 g/l (HYPO%, Siemens Advia; %HPO, Abbott), or <17 pg (%Hypo-He; Sysmex), or as a calculated transformation of the MCHC [low haemoglobin density (LHD)%, Beckman-Coulter]. It relates to iron status over the preceding 3 months. A value for %HRC of >5% suggests iron deficiency. The %HRC is affected by sample transport time and should only be used in laboratories that can reliably ensure analysis within 3–4 h of phlebotomy. This limits its usefulness to some extent, but there is cost-benefit in not having to use a reticulocyte reagent.¹⁴

Measurement of mean reticulocyte Hb content is less dependent on pre-analytical factors, although reference intervals and diagnostic thresholds are method dependent.¹⁰ There is evidence in the setting of chronic kidney disease (CKD) that these measures can predict iron response even in the face of raised serum ferritin.¹⁵ Suggested thresholds for predicting iron deficiency vary from 25 to 30 pg.^{16,17} A cut-off of 28 pg has been shown to predict iron response in paediatric patients with cancer.¹⁸ Until further data is available a threshold of 29 pg is a pragmatic recommendation.

Recommendation

- Mean reticulocyte Hb content of <29 pg can support the diagnosis of iron deficiency if initial tests are inconclusive (2B)

Table I. Summary of advanced red blood cell indices currently available.

Advanced red cell index	Parameter	Analyser
Percentage of hypochromic red cells (proportion of erythrocytes with cellular Hb <280 g/l); %HRC	%HYPO	Siemens Advia
	%HPO	Abbott
	%Hypo-He	Sysmex
	LHD%	Beckman-Coulter
Reticulocyte mean Hb content (pg)	CHr	Siemens Advia
	Ret-He	Sysmex XE/XN
	MCHr	Abbott Sapphire
	RHE	Mindray BC6800
	RHCc	ABX-Horiba Petra Nexus DX

(M)CHr, (mean) reticulocyte Hb content; Hb, haemoglobin; %HRC, percentage of hypochromic cells; LHD, low Hb density; RET-He, reticulocyte Hb equivalent; RHCc, reticulocyte Hb content calculated.

Adapted from Buttarello.¹⁰

Blood cell morphology

Morphological changes develop with iron deficient states although they are not diagnostic. In early deficiency anisocytosis precedes hypochromia and microcytosis, with more classical red cell changes (elliptocytes, pencil cells and some target cells) occurring when the Hb drops below 100–110 g/l.¹⁹

The value of inspecting a peripheral blood film in anaemia is to identify specific changes not detected by automation.¹⁰ Microscopy can suggest alternate or concurrent diagnoses including microangiopathy, myelodysplasia, thalassaemias and other haemoglobinopathies. Morphological interpretation can be affected by sample storage time, staining quality and operator skill.

Recommendation

- Inspection of a peripheral blood film should be performed in cases of diagnostic uncertainty (2C)

Serum ferritin

Ferritin represents the main bulk of protein-bound tissue iron storage. A small proportion leaks into plasma and can be quantitated accurately by immunoassay. In health there is a direct relationship between reticuloendothelial iron stores and serum ferritin,²⁰ making it a powerful laboratory test when free of confounders. Levels of <15 µg/l predict a high likelihood of iron deficiency.²¹ Levels up to 30 µg/l can still be consistent with deficiency, although is less specific.²² Reference ranges can vary by analytical method and by patient population, so local parameters should be established and regularly reassessed.^{22,23} Normal values are not significantly different between those aged <5 years compared to children aged >5 years and adults.²²

The behaviour of serum ferritin as an acute phase protein means that levels rise in response to inflammatory states as well as kidney disease, liver disease and malignancy. These variables confound clinical interpretation, a problem reflected by the difference in ranges suggested by international guidelines across a range of conditions.²⁴

Formulae for interpreting ferritin in the context of raised inflammatory markers, either C-reactive protein (CRP) or erythrocyte sedimentation rate (ESR), have been proposed but we do not consider the evidence base robust enough to apply a 'corrected' assessment of iron status in current practice.²⁵ Any clinical or laboratory evidence for an active inflammatory state should provide impetus for considering further investigation for iron deficiency regardless of a serum ferritin quantitation that lies within the normal range.^{26,27} Normal ranges will vary by laboratory; for clarity we suggest a value of <150 µg/l be used as a trigger to consider further investigation for iron deficiency.

Recommendations

- Serum ferritin of $<12 \mu\text{g/l}$ in children aged < 5 years and $<15 \mu\text{g/l}$ in those aged > 5 years is indicative of iron deficiency (1B)
- Serum ferritin within the laboratory reference range cannot exclude iron deficiency in the context of raised inflammatory markers or a history of acute or chronic disease, so further investigation may be warranted (2B)

Iron studies: serum iron, total iron binding capacity (TIBC) and transferrin saturation (TSAT)

Serum iron

Serum iron measures only the oxidised ferric form bound to transferrin and not the functional iron component of Hb. An individual with iron deficiency will usually exhibit a low concentration of serum iron, but it is a dynamic parameter with well-established day-to-day variability.²⁸ Serum iron measurement is required to allow calculation of percentage TSAT. Measuring serum iron in isolation is not helpful and its inclusion in the iron profile report to clinicians should be assessed by each individual laboratory.³

TIBC/unsaturated iron binding capacity (UIBC)/transferrin

The TIBC is calculated by adding excess iron to serum, removing unbound iron then measuring the iron concentration in the remaining sample. An alternative index, the UIBC, is more easily automated, and gives a TIBC when the value is added to the serum iron concentration.

The concentration of transferrin, the predominant plasma iron transport protein, can be measured by direct immunological assay. TIBC and transferrin rise in iron deficiency. Transferrin is a negative acute phase protein so its concentration may be reduced in inflammation. TIBC and transferrin have less variability than serum iron quantitation, but specificity remains poor.²⁹

TSAT

The TSAT is the ratio of serum iron to transferrin (or TIBC) expressed as a percentage, representing the proportion of serum transferrin binding sites occupied with iron molecules.³⁰ It relies on serum iron concentration so carries a similar caveat regarding variability and lack of specificity.²⁹ Diurnal fluctuations can be up to 70%. Clinical disorders such as malnutrition and chronic disease cause a decrease in transferrin synthesis and therefore increase the TSAT, reducing its usefulness.³¹ A diagnostic threshold for TSAT has not been well established, with some guidelines suggesting $<20\%$ as a

screening threshold. This is based on limited evidence. We prefer the increase in specificity provided by a lower target of $<16\%$, which has some support in the literature.^{32,33} It may be useful when the measurement of serum ferritin is equivocal.

Recommendation

- Serum iron studies should not be used in assessment of iron deficiency other than to calculate the TSAT (2B)
- TSAT of $<16\%$ can support the diagnosis of iron deficiency if initial tests are inconclusive (2B)

Soluble transferrin receptor (sTfR)

Developing erythroid cells adjust the number of TfRs on their surface, a proportion of which can be detected in the serum. Iron deficiency prompts an increase in receptor density and an associated rise in the sTfR concentration.³⁴

The sTfR has been suggested as an alternative to conventional laboratory tests of iron deficiency.³⁵ Braga *et al.*³⁴ compared the diagnostic accuracy of sTfR *versus* ferritin, concluding that sTfR has a potential clinical value in detecting iron deficiency, but there is a lack of robust studies defining its overall accuracy in addition to lack of standardisation and reference material.³⁶

Choi³⁷ advises that it is not a useful test in early-stage iron deficiency and is more appropriate in advanced cases, suggesting it has no advantage over ferritin. Increased levels are also seen in states of erythroid hyperplasia, such as haemolysis, megaloblastic anaemia, thalassaemia, and hypoxia.³⁸ sTfR is not useful in identifying iron deficiency in patients with beta thalassaemia trait in the early stages of deficiency but is abnormal in later stages.³⁹

Recommendation

- The sTfR assay is not recommended for investigating iron deficiency (2C)

Zinc protoporphyrin (ZPP)

Iron is incorporated into porphyrin during the final steps of haem synthesis. A reduction in available iron allows uptake of zinc instead, with ZPP produced as a measurable by-product.

Haematofluorometric measurement of ZPP has important limitations; misleading elevations are found in the presence of hyperbilirubinaemia⁴⁰ and chronic renal failure.⁴¹ A progressive spurious increase in ZPP is seen as Hb falls to $<100 \text{ g/l}$, so many subjects with moderate or severe anaemia have high ZPP irrespective of cause. These shortcomings can be overcome by washing red cells prior to testing or adjusting the Hb concentration to a standard value, but sample manipulation is time-consuming.

Even without laboratory limitations the value of ZPP as an index of iron deficiency is questionable. Raised values also

occur in functional iron deficiency, lead poisoning and conditions that stimulate excess porphyrin synthesis such as the thalassaemias.⁴² In simple iron-deficiency states, it adds no further information beyond the serum ferritin. In multifactorial anaemia it lacks specificity, as demonstrated in a population study where malaria and haemoglobinopathies were prevalent alongside iron deficiency.⁴³ The advantage of ZPP needing only a small sample volume (0.2 ml) and being an easily performed test are not enough to recommend its use in a developed country.^{44–46}

Recommendation

- Measurement of ZPP is not recommended for the diagnosis of iron deficiency (2B)

Hepcidin

The predominant form of hepcidin in humans is hepcidin-25 produced in the liver in response to iron levels, inflammation and erythropoietic demand.⁴⁷ Inflammation triggers a rise in hepcidin concentration, inhibiting iron release from macrophages, enterocytes, and hepatocytes. Conversely in iron deficiency, as iron levels fall, synthesis of hepcidin is inhibited.^{47–49} Such measurements make hepcidin a useful putative diagnostic tool to assess iron status and response to oral iron⁵⁰ although, as with other iron indices, there are confounders such as diurnal variation.⁵¹

Hepcidin assays tend to be mass spectrometry (MS) or radioimmunoassay techniques. These methods have good specificity, but their ability to detect low levels of serum hepcidin limits their sensitivity.⁵² Their use is limited in routine clinical laboratories mainly due to the expense of the equipment, availability and unsuitability for high throughput testing,⁵² and may be more appropriate in establishing a reference standard.⁵³ Work is on-going to improve standardisation material and normal ranges.⁵⁴

In recent years, enzyme-linked immunosorbent assay (ELISA)-based methods have become available. These are simple but still relatively expensive although more suitable for use in routine clinical laboratories. Very few commercial assays are currently available. However, development of the hepcidin-25 (bioactive) ELISA immunoassay has future clinical promise in a routine clinical laboratory.

Recommendation

- Measurement of hepcidin is not currently recommended for investigating iron deficiency (2C).

Summary

The present good practice paper has summarised up-to-date investigations available in the developed world for assessment

of iron deficiency in children and adults, including situations where results should be viewed with caution. It has been designed to work in conjunction with relevant specialty-specific clinical guidelines like those related to pregnancy or gastroenterological bleeding. We expect the evidence base for use of other tests such as hepcidin to evolve over the next few years allowing further development of this algorithm.

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Conflict of Interests

The BSH paid the expenses incurred during the writing of this good practice paper.

All authors have made a declaration of interests to the BSH and Task Force Chairs which may be viewed on request. The members of the writing group have no conflicts of interest to declare.

Review Process

Members of the writing group will inform the writing group Chair if any new pertinent evidence becomes available that would alter the strength of the recommendations made in this document or render it obsolete. The document will be archived and removed from the BSH current guidelines website if it becomes obsolete. If new recommendations are made an addendum will be published on the BSH guidelines website (www.b-s-h.org/guidelines).

Disclaimer

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References

1. World Health Organization (WHO). Haemoglobin concentrations for the diagnosis of anaemia and assessment of severity. Vitamin and Mineral Nutrition Information System. Geneva: World Health Organization; 2011. [Accessed: 10 June 2021] Available from: <http://www.who.int/vmnis/indicators/haemoglobin.pdf>
2. Scientific Advisory Committee on Nutrition. Iron and Health; 2010. [Accessed 10 June 2021]. Available from: <https://assets.publishing.service>

- gov.uk/government/uploads/system/uploads/attachment_data/file/339309/SACN_Iron_and_Health_Report.pdf
3. Thomas DW, Hinchliffe RF, Briggs C, Macdougall IC, Littlewood T, Cavill I. British Committee for Standards in Haematology. Guideline for the laboratory diagnosis of functional iron deficiency. *Br J Haematol*. 2013;**161**:639–48.
 4. Pavord S, Daru J, Prasannan N, Robinson S, Stanworth S, Girling J, et al. UK guidelines on the management of iron deficiency in pregnancy. *Br J Haematol*. 2020;**188**:819–30.
 5. Bain BJ, Bates I, Laffan MA. Dacie and Lewis Practical Haematology E-Book: Expert Consult: Online and Print, 12th ed. Philadelphia, PA: Elsevier Health Sciences; 2016.
 6. Lynch S. Indicators of the iron status of populations: red blood cell parameters. Assessing the Iron Status of Populations: Including Literature Reviews: Report of a Joint World Health Organization/Centers for Disease Control and Prevention Technical Consultation on the Assessment of Iron Status at the Population Level. Geneva Switzerland; 2004;Apr:6–8.
 7. Bermejo F, García-López S. A guide to diagnosis of iron deficiency and iron deficiency anemia in digestive diseases. *World J Gastroenterol*. 2009;**15**:4638–43.
 8. Lawrence AC, Bevington JM, Young M. Storage of blood and the mean corpuscular volume. *J. Clin. Pathol*. 1975;**28**:345–9.
 9. Joshi A, McVicker W, Segalla R, Favaloro E, Luu V, Vanniasinkam T. Determining the stability of complete blood count parameters in stored blood samples using the SYSMEX XE-5000 automated haematology analyzer. *Int J Lab Hematol*. 2015;**37**:705–14.
 10. Buttarello M. Laboratory diagnosis of anemia: are the old and new red cell parameters useful in classification and treatment, how? *Int J Lab Hematol*. 2016;**38**:123–32.
 11. Lippi G, Pavesi F, Bardi M, Pipitone S. Lack of harmonization of red blood cell distribution width (RDW). Evaluation of four hematological analyzers. *Clin Biochem*. 2014;**47**:1100–3.
 12. Bresani Salvi CC, Braga MC, Figueirôa JN, Batista Filho M. Could the erythrocyte indices or serum ferritin predict the therapeutic response to a trial with oral iron during pregnancy? Results from the Accuracy study for Maternal Anaemia diagnosis (AMA). *BMC Pregnancy Childbirth*. 2016;**16**:218.
 13. Ganz T. Systemic iron homeostasis. *Physiol. Rev*. 2013;**93**:1721–41.
 14. Sellors A, Melbourne J, Barratt J, Barton L. Assessing the stability and suitability of haematology parameters for diagnosing and monitoring iron deficiency. *Int J Lab Hematol*. 2017;**39**:e132–4.
 15. Padhi S, Glen J, Pordes BA, Thomas ME, Guideline Development Group. Management of anaemia in chronic kidney disease: summary of updated NICE guidance. *BMJ*. 2015;**350**:h2258.
 16. Canals C, Remacha AF, Sarda MP, Piazuelo JM, Royo MT, Romero MA. Clinical utility of the new Sysmex XE 2100 parameter-reticulocyte hemoglobin equivalent-in the diagnosis of anemia. *Haematologica*. 2005;**90**:1133–4.
 17. Chinudomwong P, Binyasing A, Trongsakul R, Paisooksantivatana K. Diagnostic performance of reticulocyte hemoglobin equivalent in assessing the iron status. *J Clin Lab Anal*. 2020;**34**:e23225.
 18. Tantawy AA, Ragab IA, Ismail EA, Ebeid FS, Al-Bshkar RM. Reticulocyte hemoglobin content (Ret He): a simple tool for evaluation of iron status in childhood cancer. *J Pediatr Hematol Oncol*. 2020;**42**:e147–51.
 19. Bain BJ. Blood cells: A Practical Guide. New York, NY: John Wiley & Sons; 2014.
 20. Walters GO, Miller FM, Worwood M. Serum ferritin concentration and iron stores in normal subjects. *J Clin Pathol* 1973;**26**:770–2.
 21. Guyatt GH, Oxman AD, Ali M, Willan A, McIlroy W, Patterson C. Laboratory diagnosis of iron-deficiency anemia. *J Gen Intern Med*. 1992;**7**:145–53.
 22. World Health Organization (WHO). Serum ferritin concentrations for the assessment of iron status and iron deficiency in populations. Vitamin and Mineral Nutrition Information System. Geneva: World Health Organization; 2011 [Accessed: 10 June 2021]. Available from: http://www.WHO.int/vmnis/indicators/serum_ferritin.pdf
 23. Horowitz GL, Altaie S, Boyd JC, Ceriotti F, Garg U, Horn P, et al. Defining, Establishing, and Verifying Reference Intervals in the Clinical Laboratory: Approved Guideline C28-A3. 3rd edn. Wayne, PA: Clinical and Laboratory Standards Institute; 2010
 24. Peyrin-Biroulet L, Williet N, Cacoub P. Guidelines on the diagnosis and treatment of iron deficiency across indications: a systematic review. *Am J Clin Nutr*. 2015;**102**:1585–94.
 25. Coenen JL, van Dieijen-Visser MP, van Pelt J, van Deursen CT, Fickers MM, van Wersch JW, et al. Measurements of serum ferritin used to predict concentrations of iron in bone marrow in anemia of chronic disease. *Clin Chem*. 1991;**37**:560–3.
 26. Cavill I, Jacobs A, Worwood M. Diagnostic methods for iron status. *Ann Clin Biochem*. 1986;**23**:168–71.
 27. Witte DL. Can serum ferritin be effectively interpreted in the presence of the acute-phase response? *Clin Chem*. 1991;**37**:484–5.
 28. Romslo I, Talstad I. Day-to-day variations in serum iron, serum iron binding capacity, serum ferritin and erythrocyte protoporphyrin concentrations in anaemic subjects. *Eur. J. Haematol*. 1988;**40**:79–82.
 29. Beard J. Indicators of the iron status of populations: free erythrocyte protoporphyrin and zinc protoporphyrin; serum and plasma iron, total iron binding capacity and transferrin saturation; and serum transferrin receptor. Assessing the iron status of populations. 2nd ed. Geneva: World Health Organization; 2007. [Accessed 10 June 2021] Available from: https://www.who.int/nutrition/publications/micronutrients/anaemia_iron_deficiency/9789241596107_annex3.pdf
 30. Coyne DW, Kapoian T, Suki W, Singh AK, Moran JE, Dahl NV, et al. DRIVE Study Group. Ferric gluconate is highly efficacious in anemic hemodialysis patients with high serum ferritin and low transferrin saturation: results of the Dialysis Patients' Response to IV Iron with Elevated Ferritin (DRIVE) Study. *J Am Soc Nephrol* 2007;**18**:975–84.
 31. Wish JB. Assessing iron status: beyond serum ferritin and transferrin saturation. *Clin J Am Soc Nephrol*. 2006;**1**(Suppl 1):S4–8.
 32. Cook JD. Diagnosis and management of iron-deficiency anaemia. *Best Pract Res Clin Haematol*. 2005;**18**:319–32.
 33. Daude S, Remen T, Chateau T, Danese S, Gastin I, Baumann C, et al. Comparative accuracy of ferritin, transferrin saturation and soluble transferrin receptor for the diagnosis of iron deficiency in inflammatory bowel disease. *Aliment Pharmacol Ther*. 2020;**51**:1087–95.
 34. Braga F, Infusino I, Dolci A, Panteghini M. Soluble transferrin receptor in complicated anemia. *Clin Chim Acta*. 2014;**20**:143–7.
 35. Punnonen K, Irjala K, Rajamäki A. Serum transferrin receptor and its ratio to serum ferritin in the diagnosis of iron deficiency. *Blood* 1997;**89**:1052–7.
 36. Brugnara C. Iron deficiency and erythropoiesis: new diagnostic approaches. *Clin Chem*. 2003;**49**:1573–8.
 37. Choi JW. Sensitivity, specificity, and predictive value of serum soluble transferrin receptor at different stages of iron deficiency. *Ann Clin Lab Sci*. 2005;**35**:435–9.
 38. Shin DH, Kim HS, Park MJ, Suh IB, Shin KS. Utility of access soluble transferrin receptor (sTfR) and sTfR/log ferritin index in diagnosing iron deficiency anemia. *Ann Clin Lab Sci*. 2015;**45**:396–402.
 39. Jalali MT, Mohseni A, Keikhaei B, Latifi M. Evaluation of diagnostic efficacy of serum sTfR assay in iron-deficiency anemia and beta-thalassemia trait in Shafa hospital, Ahvaz, Iran 2010. *Eur Rev Med Pharmacol Sci*. 2012;**16**:1441–5.
 40. Buhrmann E, Mentzer WC, Lubin BH. The influence of plasma bilirubin on zinc protoporphyrin measurement by a hematofluorimeter. *J Lab Clin Med*. 1978;**91**:710–6.
 41. Garrett S, Worwood M. Zinc protoporphyrin and iron deficient erythropoiesis. *Acta Haematol*. 1994;**91**:21–5.
 42. Graham EA, Felgenhauer J, Detter JC, Labbe RF. Elevated zinc protoporphyrin associated with thalassemia trait and hemoglobin E. *J Pediatr*. 1996;**129**:105–10.
 43. Asobayire FS, Adou P, Davidsson L, Cook JD, Hurrell RF. Prevalence of iron deficiency with and without concurrent anemia in population groups

- with high prevalences of malaria and other infections: a study in Cote d'Ivoire. *Am J Clin Nutr.* 2001;**74**:776–82.
44. Gammon RR, Kozel T, Morel P, Kendrick C. Evaluation of iron stores by zinc protoporphyrin analysis in blood donors. *Lab Med.* 2018;**49**:311–5.
 45. Mei Z, Flores-Ayala RC, Grummer-Strawn LM, Brittenham GM. Is erythrocyte protoporphyrin a better single screening test for iron deficiency compared to hemoglobin or mean cell volume in children and women? *Nutrients.* 2017;**9**:557.
 46. Homann C, Hennig G, Maier F, Stepp H, Holdt LM, Vogeser M, et al. Non-invasive measurement of erythrocyte zinc protoporphyrin in children. *Pediatr Res.* 2019;**85**:349–54.
 47. Cheng PP, Jiao XY, Wang XH, Lin JH, Cai YM. Hepcidin expression in anemia of chronic disease and concomitant iron-deficiency anemia. *Clin Exp Med.* 2011;**11**:33–42.
 48. Ganz T. Hepcidin – a regulator of intestinal iron absorption and iron recycling by macrophages. *Best Pract Res Clin Haematol.* 2005;**18**:171–82.
 49. Ganz T, Nemeth E. Iron sequestration and anemia of inflammation. *Semin Hematol.* 2009;**46**:387–93.
 50. Girelli D, Nemeth E, Swinkels DW. Hepcidin in the diagnosis of iron disorders. *Blood.* 2016;**127**:2809–13.
 51. Kroot JJ, Hendricks JC, Laarakkers CM, Klaver SM, Kemna EH, Tjalsma H, et al. (Pre)analytical imprecision between subject variability, and daily variations in serum and urine hepcidin: implications for clinical studies. *Anal Biochem.* 2009;**389**:124–9.
 52. Young MF, Glahn RP, Ariza-Nieto M, Inglis J, Olbina G, Westerman M, et al. Serum hepcidin is significantly associated with iron absorption from food and supplemental sources in healthy young women. *Am J Clin Nutr.* 2009;**89**:533–8.
 53. Murphy AT, Witcher DR, Luan P, Wroblewski VJ. Quantitation of hepcidin from human and mouse serum using liquid chromatography tandem mass spectrometry. *Blood.* 2007;**110**:1048–54.
 54. Diepeveen LE, Laarakkers CM, Martos G, Pawlak ME, Uğuz FF, Verberne KE, et al. Provisional standardization of hepcidin assays: creating a traceability chain with a primary reference material, candidate reference method and a commutable secondary reference material. *Clin Chem Lab Med.* 2019;**57**:864–72.